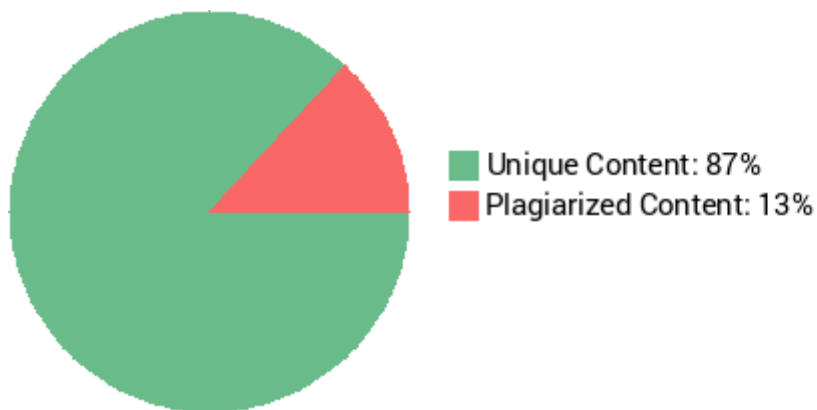




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Smart Grid Sentinel: Comparative Time-Series Forecasting for Load Shedding Risk Prediction and Intelligent Grid Alerts

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Developing power grids frequently struggle with load shedding due to fragile infrastructures. In this paper, we introduce Smart Grid Sentinel, a forecasting framework designed to predict grid instability and localized load-shedding risks at the Upazila level. We train and evaluate twelve models, ranging from traditional machine learning methods to sequence-to-sequence neural networks and time-series Transformers, using a dataset of 65,983 telemetry records from 41 Upazilas in Sylhet and Chattogram, Bangladesh. To ensure realistic out-of-sample evaluation and prevent temporal leakage, we split the data chronologically (80/20) and fit scaling parameters exclusively on the training partition. Our findings show that the Informer architecture achieves a Weighted F1 score of 88.44% and a classification accuracy of 87.41%. While a standard Gated Recurrent Unit (GRU) achieves a slightly higher raw accuracy of 87.69%, it experiences severe validation loss divergence (overfitting) after 20 epochs. In contrast, the Informer shows stable training, handles class imbalance effectively (79.00% recall on the minority Medium-Risk class), and processes predictions rapidly. To make these predictions actionable, we implement a rule-based explainability module that translates high-risk classification outputs into clear operational protocols for grid operators and citizens.

Smart Grid, Load Shedding Prediction, Time Series Forecasting, Informer, Transformer, Bangladesh.

Introduction

Managing electrical load imbalances requires predictive rather than reactive strategies to prevent sudden outages and load shedding. In developing power grids like that of Bangladesh, operators regularly face load fluctuations driven by erratic demand, local weather changes, and deteriorating substation hardware. Existing distribution systems usually operate reactively, offering no early warnings to forecast local load shedding risk in advance for scheduling.

In this work, we present a time-series forecasting pipeline to model localized load shedding risk for power distribution grids. We establish a 2-hour-ahead sequence prediction setup using 10-hour historical telemetry windows. Using this setup, we train and evaluate twelve models: traditional classifiers (Logistic Regression, Decision Trees, Random Forests, and XGBoost), recurrent neural networks (LSTMs, GRUs, and CNN-LSTMs), and time-series Transformers (Autoformer, Informer, PatchTST, Temporal Fusion Transformer, and Times Net).

Additionally, deep learning networks are often seen as black-box models, which makes grid operators hesitant to deploy them in practice. To overcome this limitation, we develop a rule-based explainability layer. This layer translates numerical risk predictions into natural language notifications, translating complex sequential predictions into actionable operational advice.

Key Contributions

The main contributions of this paper are:

- We design a localized load shedding risk forecasting pipeline configured at the Upazila level for regional grids in Bangladesh.
- We perform a systematic benchmark of twelve machine learning, recurrent, and transformer architectures under a leakage-free, chronological evaluation split.
- We justify and validate the Informer model for production deployment, considering its training stability, classification metrics, and computational efficiency.
- We integrate a rule-based text explanation module to provide operational context for high-risk alerts.

Literature Review

Standard grid load forecasting and stability models have historically used linear statistical approaches, such as AutoRegressive Integrated Moving Average (ARIMA). Although ARIMA models capture trends in stationary, single-variable series, they fail to model complex, non-linear relationships between meteorological changes and power demand dynamics. Classical machine learning classifiers, such as Random Forests and XGBoost, offer better performance by handling high-dimensional inputs. However, these tabular methods treat observations as independent, ignoring sequential patterns and temporal dependency.

To capture sequential relationships, recurrent neural networks like Long Short-Term Memory (LSTM) [6] and Gated Recurrent Units (GRU) [7] are frequently applied. These networks use persistent hidden states to carry sequential information across timestamps. Additionally, hybrid architectures like CNN-LSTMs [10] combine 1D

convolutional operations for spatial feature extraction with recurrent nodes for sequence forecasting.

More recently, deep self-attention models have set new baselines in sequence modeling. The Informer [1] uses a ProbSparse self-attention mechanism to capture long-range dependencies with lower computational complexity, whereas PatchTST [3] breaks sequences into overlapping sub-series patches to preserve local correlations. However, current research seldom focuses on localized, Upazila-level load shedding forecasting, especially in areas with high grid instability, and rarely integrates machine learning predictions with rule-based text explanation layers.

[Table: Summary of Related Works and Proposed Work]

Study	Method	Dataset	Key Limitation
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Rashid et al. [12]	LSTM	Smart Grid	No explainability layer
Chen et al. [8]	XGBoost	Load Forecasting	No localized prediction
Zhou et al. [1]	Informer	Energy Data	No alert system
Proposed	Informer + Rule	Bangladesh Grid	Limited to two divisions

Dataset and Preprocessing

Data Source and Characteristics

Our experimental telemetry data includes 65,983 records collected from 41 Upazilas in the Sylhet and Chattogram divisions of Bangladesh. The dataset contains 27 initial variables, which capture weather measurements (temperature, wind speed, rainfall, humidity), timestamp features, and operational grid parameters (substation identity, electrical demand, localized transformer load, and renewable supply). A summary of these statistics is presented in Table I.

[Table: Dataset Statistics Summary]

Attribute	Value
---	---
Total Initial Records	65,983
Post-Shifting Valid Records	64,983
Number of Upazilas	41
Feeder Channels	1,000
Number of Features	27

| Observation Interval | 2 Hours |
| Forecast Horizon | 2 Hours (1 Step) |

Feature Engineering

We generated several domain-specific indicators to capture electrical and environmental stress on the local grid:

- Load Utilization Ratio (U_{load}): This measures the current load on a transformer relative to its total capacity:

Equation: $U_{load} = (\text{Transformer Load} / \text{Transformer Capacity} + 10^{-5})$

- Demand Utilization Ratio (U_{demand}): This indicates total power demand scaled by the transformer's capacity limit:

Equation: $U_{demand} = (\text{Electricity Demand} / \text{Transformer Capacity} + 10^{-5})$

- Temperature-Humidity Index (THI): This metric assesses environmental heat stress, which degrades transformer capacity and operational efficiency:

Equation: $THI = T + 0.55 (1 - H/100) (T - 14.5)$

where T represents temperature in $^{\circ}\text{C}$ and H represents relative humidity (%).

- Interaction and Peak Features: We constructed an interaction term between temperature and wind speed, alongside binary flags representing weekend periods and daily peak load intervals (18:00-22:00).

Temporal Leakage Prevention and Scaling

To build a rigorous and unbiased model validation framework, we apply the following processing constraints:

- Target Shifting: To model a 2-hour-ahead forecasting horizon, we shift the target risk class ('risk_level') by -1 index. Predictions at time $t+1$ rely exclusively on inputs available at time t .
- Temporal Partitioning: Data is sorted chronologically and grouped by 'substation_id' and 'feeder_id'. Within each feeder channel, the first 80% of observations form the training set, while the remaining 20% constitute the test set. Shuffling is disabled.
- Data Alignment: Shifting the target label by -1 introduces a single trailing null value (NaN) for each of our 1,000 distinct feeder lines. Dropping these rows reduces our dataset size from 65,983 to 64,983 observations. The resulting clean dataset is partitioned chronologically into 51,587 training records (80%) and 13,396 test records (20%).
- Sequence Padding and Truncation: Sequence-based models use a sliding window of length 5 (representing 10 hours of grid history). For each of the 1,000 feeders, the first 4 timestamps in the test partition cannot form a complete sequence. Omitting these incomplete windows leaves exactly 10,027 valid test sequences for evaluation.
- Pipeline Isolation: We fit scaling parameters (using 'Standard Scaler') solely on the training set. We then transform the test set using these pre-fitted parameters, avoiding lookahead leakage.

We explicitly avoid random shuffling and K-fold cross-validation because of temporal dependencies in electrical time-series data. Evaluating models chronologically mimics real-world operations, where future inputs are unavailable, and prevents overoptimistic validation metrics.

Proposed Methodology

We present the architecture of the Smart Grid Sentinel framework in Fig. 1, which works as a sequential forecasting pipeline.

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